Models Know, But Don't Always Say: A Guided Reproduction of Two Results on Hidden Knowledge in Language Models

Jiayu Liao **Zeyu Pan** **Yuhan Sun** **Kristian Tokos**
Group 8 CSI-435/535 Artificial Intelligence, Fall 2026
`course project reproduction report`

Abstract

A language model's output need not reflect what it internally represents. We reproduce two published results that make this gap concrete. First, following Marks & Tegmark (COLM 2024), we show that LLaMA-2-13B represents the truth of simple factual statements as a linearly accessible direction: true and false statements separate into two clusters under PCA, and linear probes trained on one dataset of statements generalize to others—while the tested TF–IDF baselines generally stay near chance, supporting signal beyond these lexical features. Second, following Cywiński et al. (2025), we study *Taboo* models—Gemma-2-9B models fine-tuned to use a secret word downstream while never uttering it—and read the secret out of intermediate layers with the logit lens, and turn those per-layer reads into a generation-free *monitor* that flags hidden knowledge without ever consulting the output text. The core results reproduce published methods using public code and models; we add targeted transfer analyses and text controls. Frozen-model inference provides the activations, and small probes are trained separately. The running theme: *the model knows, but does not necessarily say*—and simple tools from an introductory AI course suffice to see it.

1 Introduction

When a language model asserts a falsehood fluently, does any part of it "know" better? And when a model is explicitly trained to hide a piece of knowledge, does the knowledge stay hidden on the inside? Both questions ask about the gap between what a model *says* and what it *represents*—a gap with direct consequences for AI safety, where monitors that read only a model's text output can miss what the model is actually computing.

This report reproduces two published answers, chosen because they are striking, complementary, and achievable with the tools of an introductory AI course (PCA, logistic regression, linear discriminant analysis):

- **Reproduction A (main).** Marks & Tegmark [Marks and Tegmark, 2024] showed that LLaMA-2 models represent the truth of simple factual statements along linear directions: true and false statements cluster separately under PCA, probes generalize across datasets, and adding the "truth direction" to activations causally flips the model's answers.
- **Reproduction B (secondary).** Cywiński et al. [Cywiński et al., 2025] released *Taboo* models that know a secret word, use it, but never say it. White-box methods (logit lens on intermediate layers) recover the secret far more often than black-box prompting.

We re-run both pipelines end-to-end—activation extraction, probing, and interventions for A; logit-lens extraction for B—on the authors' public code and models, and report where our numbers match

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

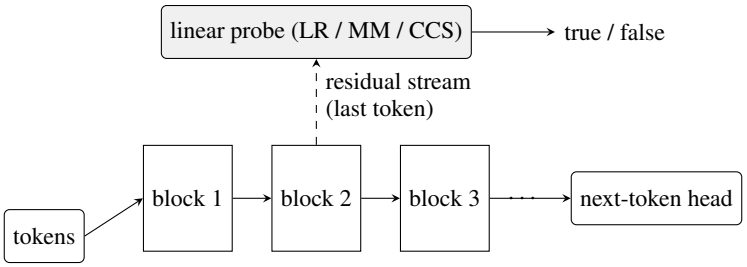


Figure 1: **Reading the residual stream.** After each transformer block, every token position carries a d-dimensional vector. A linear probe trained on these vectors tests whether a property (e.g. truth) is linearly represented; the logit lens instead decodes intermediate states directly into vocabulary space.

the originals. The core reproduction results are linked to the original paper's figures or tables; targeted failure analyses and text controls are our extensions.

2 Background: reading activations

A decoder-only transformer [Touvron et al., 2023, Gemma Team, 2024] maintains, after each block, a *residual stream*: one d-dimensional vector per token position ($d = 5120$ for LLaMA-2-13B). If a property of the input (e.g. truthfulness) is linearly decodable from these vectors, we say the model *linearly represents* it [Alain and Bengio, 2017].

Probes. We use the three probe families of Marks and Tegmark [2024], all bias-free on mean-centered activations: (i) **LR**, logistic regression trained with AdamW; (ii) **MM** (mass-mean), the closed-form mean-difference direction $\theta = \mu^+ - \mu^-$, whose covariance-whitened variant is Fisher's linear discriminant; (iii) **CCS** [Burns et al., 2023], a contrastive probe with an unlabeled optimization objective. For a statement/negation pair, write $p = p_\theta(x)$ and $q = p_\theta(\neg x)$. CCS minimizes $\mathrm{E}[(p + q - 1)^2 + \min(p, q)^2]$: consistency plus confidence. The second term discourages the constant-0.5 solution. Our implementation then uses training labels to resolve the direction's sign ambiguity; it is not an entirely label-free monitoring pipeline.

Logit lens. The logit lens [nostalgebraist, 2020] applies the model's final layer norm and unembedding to intermediate residual streams, reading off which vocabulary tokens the intermediate state already "prefers". We use it to ask, layer by layer: *where in the network does the hidden knowledge become visible?*

3 Reproduction A: the geometry of truth

3.1 Setup

We follow Marks and Tegmark [2024] (arXiv:2310.06824v3; COLM 2024) and their public repository. **Model**: LLaMA-2-13B [Touvron et al., 2023], bfloat16, inference only. **Datasets**: the repository's true/false statement datasets—e.g. `cities` ("The city of c is in C.", 1,496 statements), `neg_cities` (negated variant), `sp_en_trans` (Spanish–English translations), and eight others with fixed templates and known ground truth. **Protocol**: raw declarative statements (no QA formatting, no few-shot priming); we extract the residual stream at the *final token* (the sentence-ending period) at 12 layers spanning the model, with per-dataset mean centering; probes are trained on a random 80/20 split (seed 0). Our activation extractor is a plain-`transformers` reimplementation that writes the exact file layout of the original repository (the upstream script depends on an outdated `nnsight` API); the probe training code is the authors' own.

Table 1: True/false statement datasets of Reproduction A [Marks and Tegmark, 2024]. The first eight are curated with fixed templates; the last three are uncurated. All have known ground-truth labels.

Dataset	Statement template	n
`cities`	The city of c is in C.	1,496
`neg_cities`	The city of c is not in C.	1,496
`sp_en_trans`	The Spanish word 'x' means 'y'.	354
`neg_sp_en_trans`	... does not mean ...	354
`larger_than`	x is larger than y.	1,980
`smaller_than`	x is smaller than y.	1,980
`cities_cities_conj`	It is the case both that s_1 and that s_2.	1,498
`cities_cities_disj`	It is the case either that s_1 or that s_2.	1,498
`companies_true_false`	company/industry claims [Azaria and Mitchell, 2023]	1,200
`common_claim_true_false`	natural claims (CommonClaim)	4,450
`counterfact_true_false`	counterfactual statements (Counterfact)	31,964

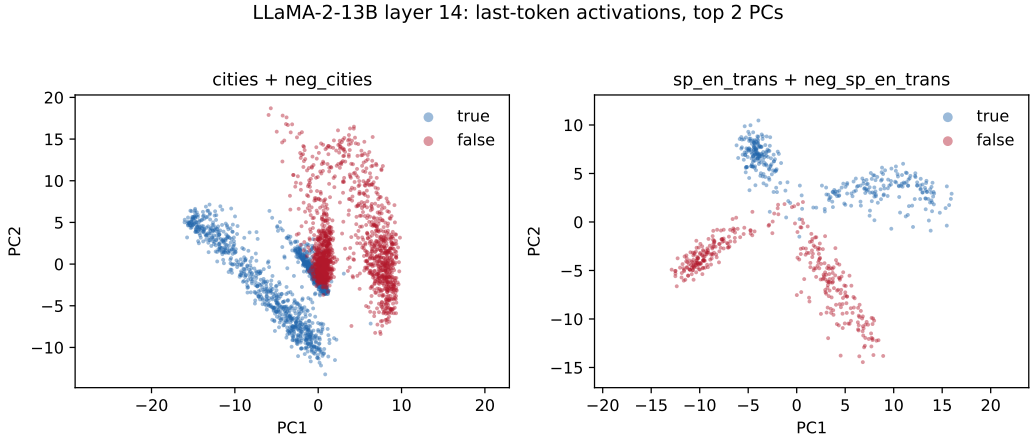


Figure 2: **Two clusters without any labels.** Each point is one statement's last-token activation projected onto the top two principal components (PCA is unsupervised). True and false statements separate geometrically. Reproduces the Fig. 1 analysis of Marks and Tegmark [2024].

3.2 Result 1: true and false statements separate under PCA

Figure 2 shows the headline phenomenon: without ever seeing labels, PCA on mid-layer activations separates true from false statements into two clusters. Truth is therefore *linearly accessible* in the representation.

3.3 Result 2: probes generalize across datasets

We trained LR, MM, and CCS probes on one (combination of) dataset(s) and evaluated on all eleven (Fig. 3), reproducing the generalization matrix of Fig. 5 of the original paper. Our reproduction confirms the original's landmarks: probes trained on `larger_than+smaller_than` reach 97% on `sp_en_trans` (LR and MM alike; original: >95%); and the transfer failures cluster exactly where the original's do — on negated statements and across comparison direction (analyzed in §3.4). Truth is also most readable in the middle of the network: held-out accuracy saturates from layer ∼10 onward, while transfer to other datasets peaks at mid-stack and decays toward the final layers (Fig. 4), consistent with the original's choice of mid-layer probes.

3.4 Analysis: negation breaks transfer — and unsupervised probes fix it

The matrix hides a clean pattern (Fig. 5). Probes trained on affirmative statements alone collapse exactly on the negated variants: LR trained on `cities` drops from 0.99 to **0.33** on `neg_cities` and

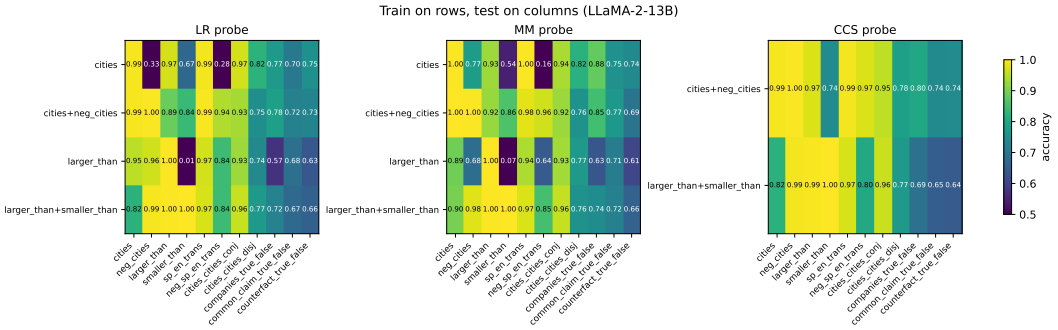


Figure 3: **Do truth probes generalize?** Each row trains on one (combination of) dataset(s); each column tests on another. Reproduces the Fig. 5 analysis of Marks and Tegmark [2024].

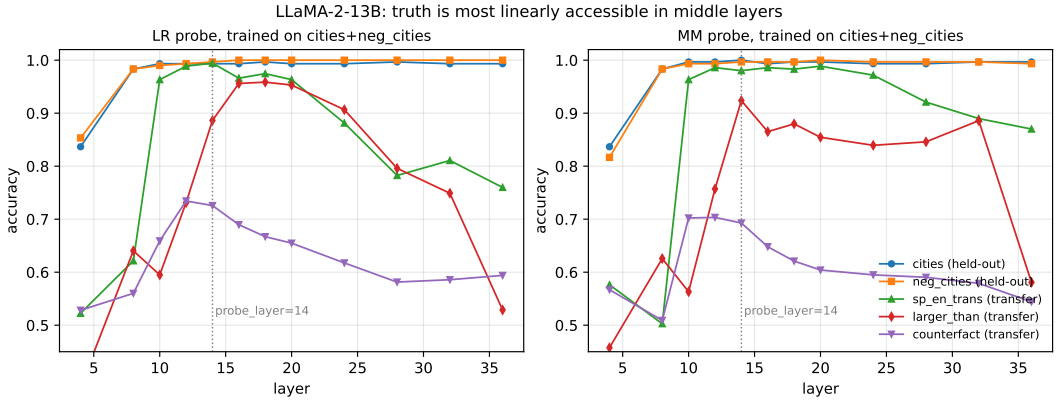


Figure 4: **Truth is most linearly accessible in middle layers.** Accuracy of LR and MM probes trained on `cities+neg_cities` at each layer, evaluated held-out (blue/orange) and transferred to other datasets (green/red/purple). Dotted line: the original repository's probe layer (14).

to **0.28** on `neg_sp_en_trans`; the MM probe is more robust within the same template family (0.77 on `neg_cities`) but collapses just as hard away from it (0.16 on `neg_sp_en_trans`). Two independent repairs work. Adding the negated counterparts to the training set restores 0.94–1.00 on the four datasets shown in Fig. 5. CCS uses the same statement/negation pairs, optimizing consistency and confidence without truth labels before using training labels to align the sign. It broadly tracks supervised probes (Fig. 6) and reaches 0.97 on `neg_sp_en_trans`. These results are consistent with template-dependent truth representations; this analysis does not directly measure a geometric rotation or establish what the model believes.

3.5 Control: comparison with lexical text baselines

Could a classifier that reads only the statement strings — never touching the model — do just as well? We trained TF–IDF + logistic regression on the raw text under the identical 80/20 split (word 1–2-grams; a stronger char 3–5-gram variant as well). Figure 7: the baseline sits at chance (0.45–0.52) on 10 of 11 datasets, including every negated and cross-lingual variant, and char n-grams do no better (≤ 0.55 everywhere). The lone exception is `larger_than` (0.94), where digit tokens leak magnitude — and the same baseline flips to 0.05 on `smaller_than`, showing it learned the surface direction, not the relation. These datasets are built so that every subject and every predicate token appears in both classes; only the *pairing* carries the truth value, and the pairing is what these TF–IDF features do not reliably capture. Stronger text models may perform differently; this control does not show that text lacks truth information.

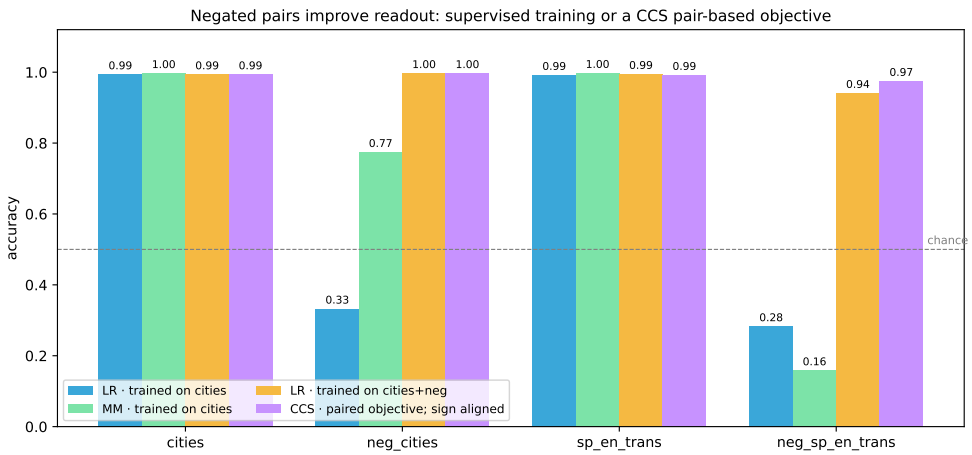


Figure 5: **The transfer failures live on negated statements.** Probes trained on affirmative statements alone (blue/green) collapse exactly on the negated variants; adding negated counterparts to training (orange) repairs it; CCS (purple) uses a pair-based objective without truth labels during optimization, followed by training-label sign alignment. Same LLaMA-2-13B layer-14 protocol as Fig. 3.

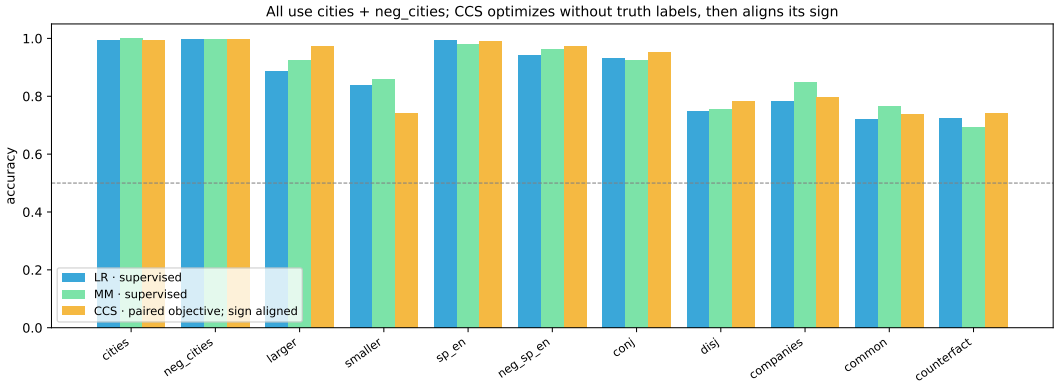


Figure 6: **Learning from contrast pairs.** CCS optimizes consistency and confidence without truth labels, then uses training labels to align its sign. We compare it with supervised LR and MM probes across all eleven datasets; all three are trained on `cities+neg_cities`.

3.6 Result 3: pushing along the truth direction flips answers

Following §6 and Table 2 of the original (targets reproduced in Table 2): we take the MM direction trained on `cities+neg_cities`, normalized so that adding it moves the mean false statement onto the mean true one from the probe's point of view, add it to the residual stream at layers 8–14 at the final tokens, and measure $P(\text{TRUE}) - P(\text{FALSE})$ under a few-shot TRUE/FALSE prompt on held-out `sp_en_trans` statements. **The sign of the gap flips in both directions** (Fig. 8): true statements drop from $+0.12$ to -0.09 when the direction is subtracted, and false statements rise from -0.09 to $+0.06$ when it is added. The effect is directionally exactly as the original reports; our effect sizes are smaller than the original's NIE values, which we attribute to our single fixed steering scale — the causality claim itself (*the probe direction is used by the model*) reproduces clearly, and the original's central warning stands: *classification accuracy does not imply causal use*.

4 Reproduction B: reading a secret the model never says

Cywiński et al. [2025] trained *Taboo* models: LoRA adapters on `gemma-2-9b-it` [Gemma Team, 2024] that play a taboo game—they know a secret word (e.g. *gold*, *moon*, *cloud*),

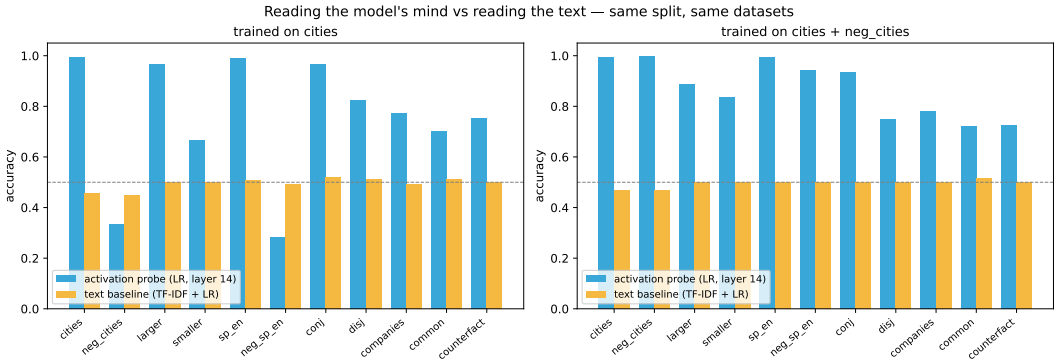


Figure 7: **Reading the text vs. reading the model.** A TF–IDF + logistic-regression baseline on the raw statement strings (amber) sits at chance on 10 of 11 datasets under the identical split, while the layer-14 activation probe (blue) separates the same statements. This supports signal beyond these tested lexical features, not the failure of all text models.

Table 2: Causal-intervention targets from Marks and Tegmark [2024] (their Table 2, LLaMA-2-13B): normalized indirect effect (NIE; 0 = no effect, 1 = full flip) of adding the probe direction to the residual stream. Note the gap: LR and MM classify equally well, but MM directions are far more causal.

Probe direction (13B)	NIE false→true	NIE true→false
LR, trained on `cities`	0.13	0.19
MM, trained on `cities`	0.77	0.90
MM, trained on `cities+neg_cities`	0.85	0.97
CCS, trained on `cities+neg_cities`	0.31	0.73

use it to answer hint questions, and never utter it. We load the three released adapters (`bcywinski/gemma-2-9b-it-taboo-{gold,moon,cloud}`), prompt each model with eight hint questions, and apply the logit lens (Fig. 9) at every layer, reading the rank of the secret word in each layer's next-token distribution at the assistant control-token positions (Fig. 10).

The secret is readable mid-stack, and only in the fine-tuned models. For *moon*, the secret word enters the top-5 logits on all eight prompts at layers 7–15, with mean rank $= 1$ at layer 11; for *gold*, top-5 on 62% of prompts at layers 7–10; *cloud* is weaker and later (top-5 at 62% around layer 29). The base model without the adapter never places the word in the top-5 at any layer (mean rank $\sim$ 4,000–6,000). This mirrors the original's central claim: the secret is hidden from the output, but not from the activations. Table 3 lists the original results we compare against: white-box access lifts an auditor's top-1 accuracy on Taboo from 42% (text only) to 84% (logit-lens tokens), while the base model shows no such signal (19%); black-box attacks on Taboo stay below 8%.

From reading to monitoring. The per-layer ranks collapse into a one-number *monitor score*: the share of the 42 layers at which the secret word sits in the model's top-5 predictions at the control-token positions. We flag a conversation as HIDDEN INTENT DETECTED when the score reaches 25%, or when any sustained run of ≥ 4 consecutive top-5 layers occurs. Applied to our logs, the monitor flags *moon* on $8/8$ prompts, *gold* on $5/8$, and *cloud* on $3/8$ (weaker and later, peak rank 1 at layer 29), and never fires on the base model ($0/8$). Note what the monitor does *not* read: the output text. The entire judgment is computed from hidden states—a small, generation-free classifier with a confidence score, watching the layers that §3.3 identified as most readable. We demo this monitor as an interactive offline page in the final presentation.

5 Discussion and limitations

The core results reproduce published work; our additional transfer analyses and text controls distinguish observed readout from broader claims about knowledge. These experiments use course-level tools and frozen-model inference, with small probes trained on the resulting activations. Four limits

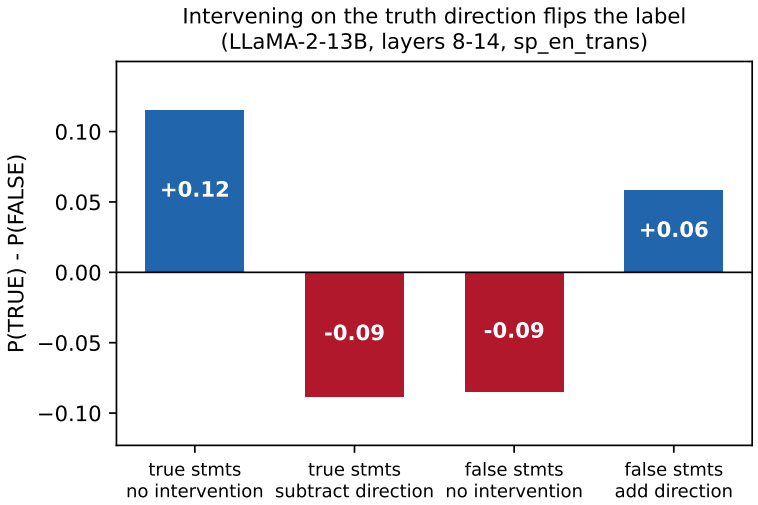


Figure 8: **The probe direction is causally used.** Adding the MM truth direction to the residual stream (layers 8–14) moves $P(\text{TRUE}) - P(\text{FALSE})$ for *false* statements from negative to positive; subtracting it does the reverse for *true* statements (177 `sp_en_trans` statements per condition, few-shot TRUE/FALSE prompt). Compare Table 2 of Marks and Tegmark [2024].

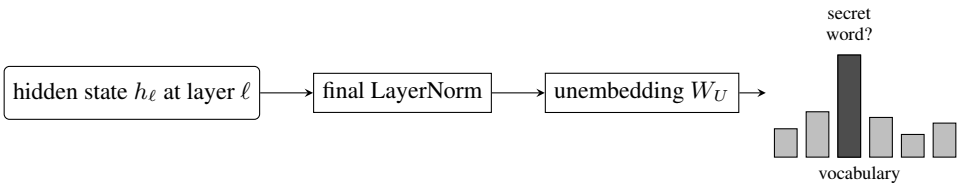


Figure 9: **Logit lens.** Decoding an intermediate hidden state with the model's own final layer norm and unembedding reveals which tokens the layer "prefers". For Taboo models, the secret word rises to the top ranks at middle layers even though the output never contains it.

bound what the results buy us. First, **white-box only**: every method here reads weights and activations, so none of them transfer to closed API models. Second, **template statements**: the datasets are simple and template-based; natural lies are messier. The transfer failures are consistent with template-dependent representations (§3.4); this analysis does not directly measure angles between directions. Third, **one model, one scale**: LLaMA-2-13B and Gemma-2-9B; other families and sizes may draw the geometry differently. Fourth, **steering, not yet control**: the intervention (§3.6) closes the classification-vs-causation gap in one setting, with a single fixed steering scale and smaller effect sizes than the original; a monitor that *reads* is demonstrated, a monitor that *corrects* is not.

Where this points. The reproduction motivates small readers of internal states, while leaving substantial questions open. CCS shows that contrast structure can guide optimization without truth labels, but our implementation still needs labels for sign alignment. The layer sweep identifies useful locations in these models, not a universal monitoring layer. The Taboo experiment reads a controlled secret; it does not establish belief or deceptive intent. Our subsequent EM behaviour study, presented separately in the midterm deck, tests ordinary questions, saved generations, and activation probes. It needs fresh prospective evaluation and better calibration before any deployment claim. Reading a signal, using a reliable threshold, and establishing intent are different experimental goals.

Table 3: Original results from Cywiński et al. [2025] (their Fig. 5–6 and `METRICS.md`) that we compare against: auditor top-1 accuracy on the Taboo secret word.

Elicitation method	auditor top-1 (%)
Text output only, standard prompts (I/O baseline)	42.1
+ logit-lens tokens (white-box)	84.0
+ activation-similarity tokens (white-box)	85.1
Logit lens on the *base* model (control)	19.4
Best black-box attack, adversarial prompts	7.2

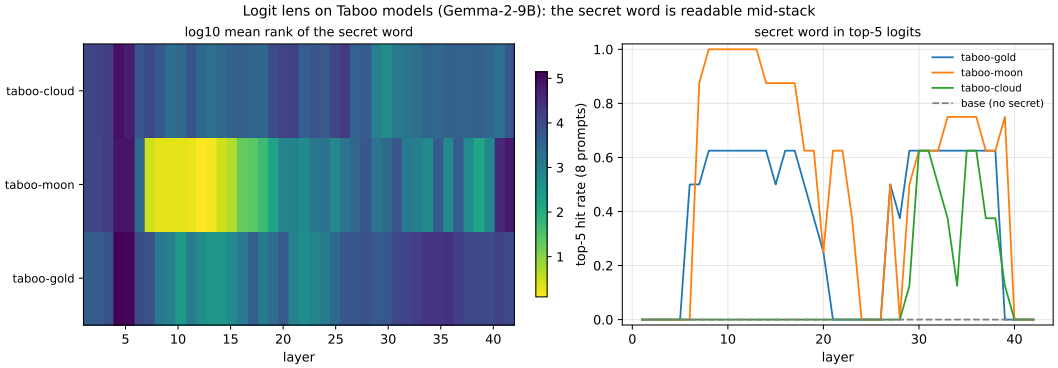


Figure 10: **Where the secret lives.** Left: $\log_{10}$ mean rank of the secret word at each layer (brighter = more readable). Right: fraction of the eight hint prompts for which the secret word is in the layer's top-5 logits. The base model control (dashed) never contains the signal. Our reproduction of the logit-lens analysis of Cywiński et al. [2025].

Acknowledgments and Disclosure of Funding

This report reproduces published results: Reproduction A follows Marks and Tegmark [2024] (code: `https://github.com/saprmarks/geometry-of-truth`); Reproduction B follows Cywiński et al. [2025] (code and models: `https://github.com/cywinski/eliciting-secret-knowledge`). All original credit belongs to those authors; any errors in the reproduction are ours.

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